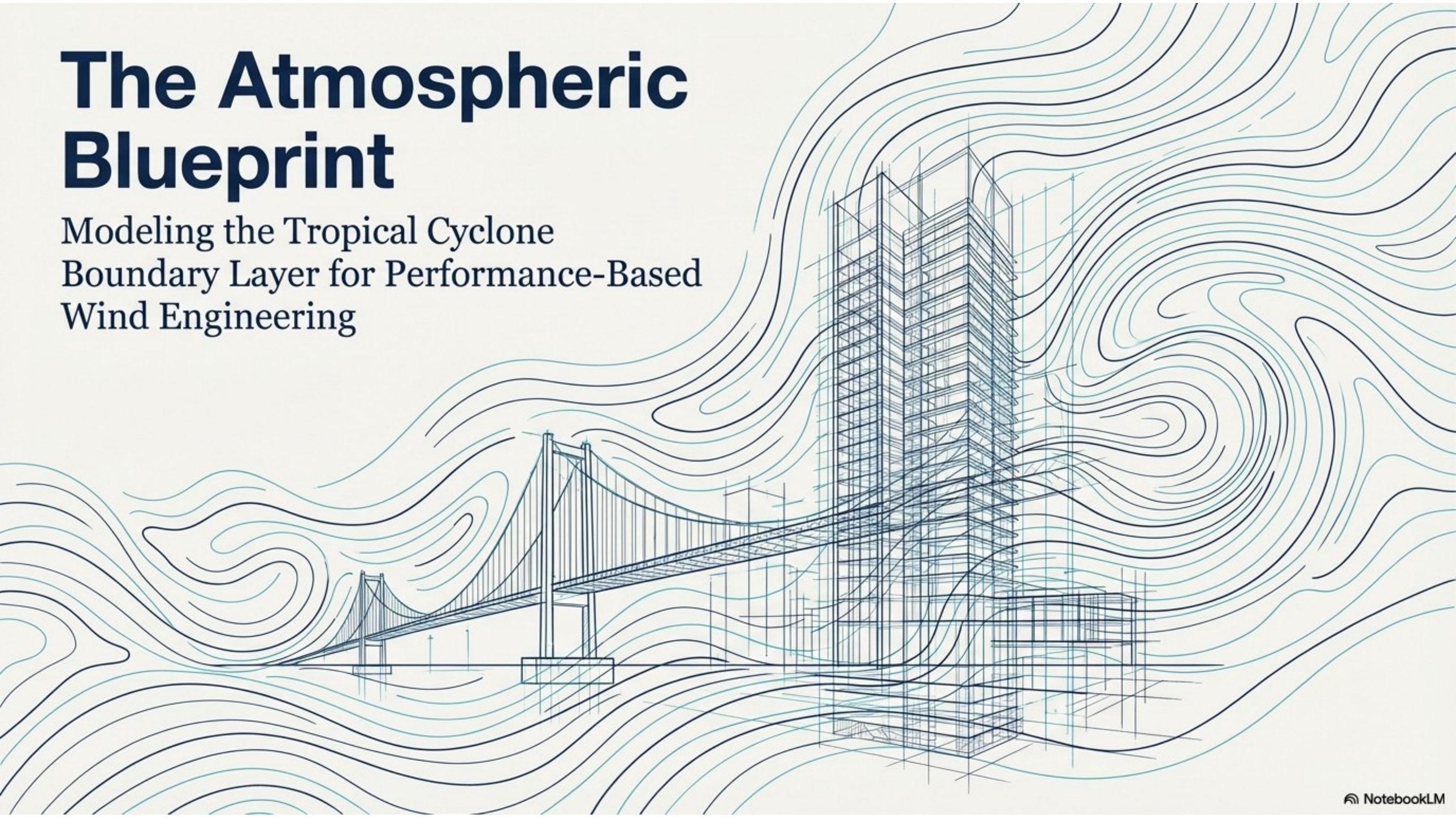
The Atmospheric Blueprint

Modeling the Tropical Cyclone Boundary Layer for Performance-Based Wind Engineering



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Engineering for **extreme winds requires probabilistic foresight**

The PBWE Chain

Hazard
λ(IM)
Tropical Cyclone extreme wind models (The focus of this deck).

→

Interaction
f(IP|IM,SP)
Wind pressures and forces on the structure.

→

Demand
f(EDP|IP,SP,IM)
Peak dynamic responses and debris impact.

→

Damage
f(DM|EDP)
Envelope breaches, fatigue, water ingress.

→

Performance
G(DV|DM)
Systematic fragility and total financial/human loss.

Key Insight: To accurately predict structural failure (like Hurricane Katrina's $125B in losses), engineers run Monte Carlo simulations generating thousands of artificial storms. The foundational input for this entire chain is the simulated wind hazard.

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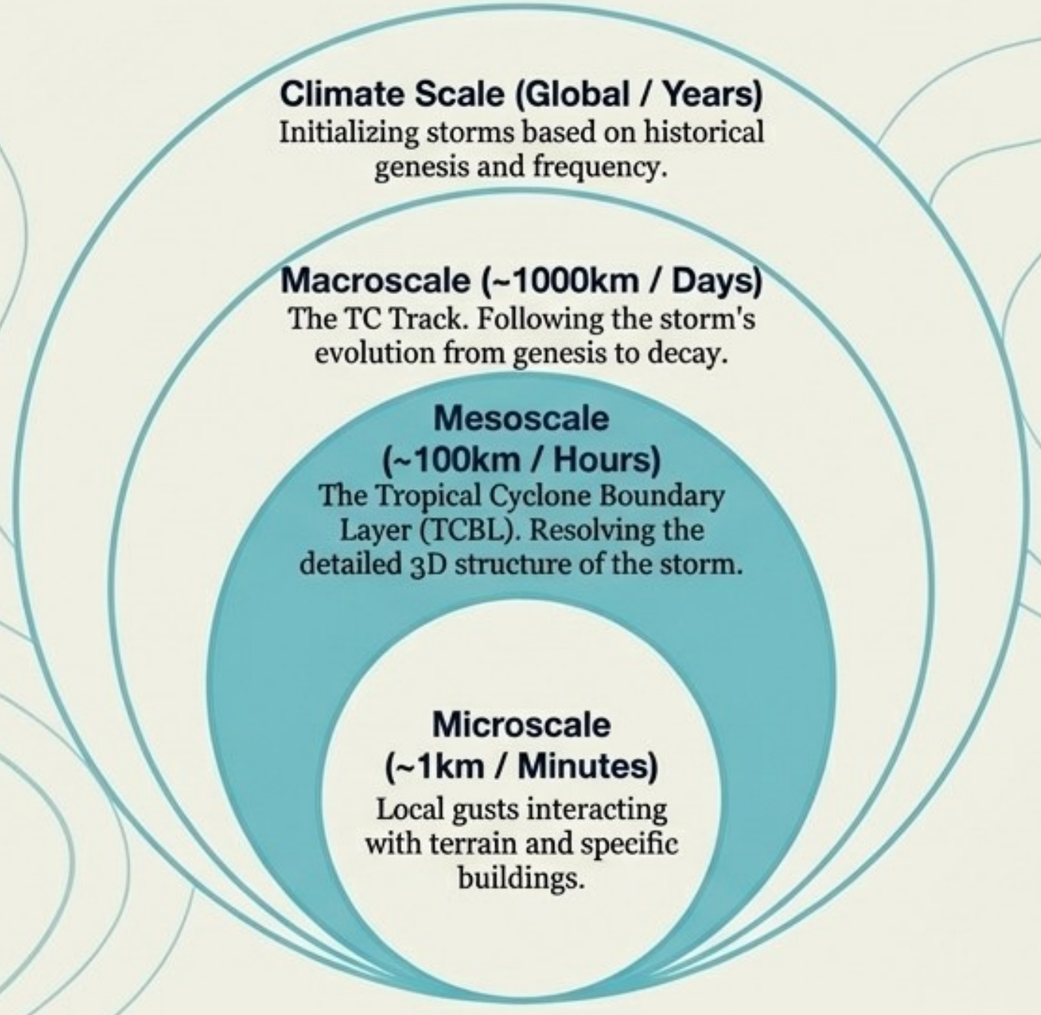
Simulating a storm requires nesting scales from global to local



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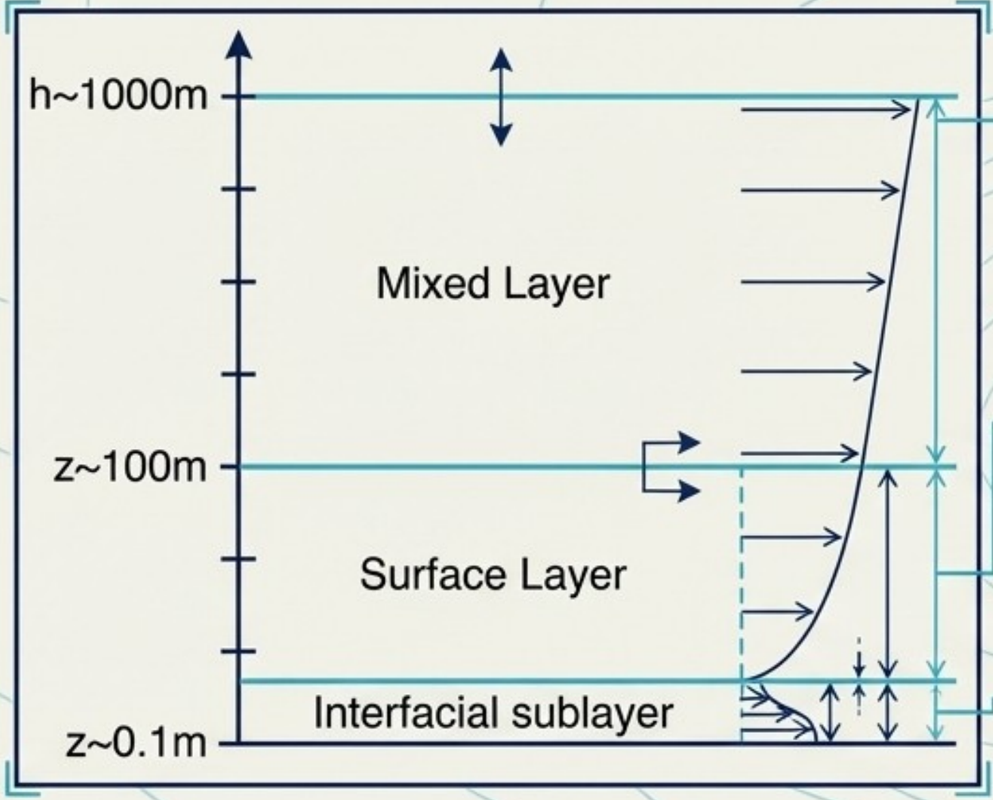
The physical arena: Anatomy of the Tropical Cyclone Boundary Layer



Mixed Layer (~200m to 1000m)

Surface influence diminishes. Capped by the free atmosphere.

Surface Layer (~200m)

The flux-constant layer. Shear-generated turbulence drives diffusivity; heat/momentum fluxes remain constant. Logarithmic profile.

Interfacial Sublayer (~0.1m)

Molecular constraints dominate momentum exchange.

Key Insight: The TCBL is defined dynamically—often marked by the height of maximum angular momentum, which rises from ~200m near the storm's eye to 1200m on the outer edges.

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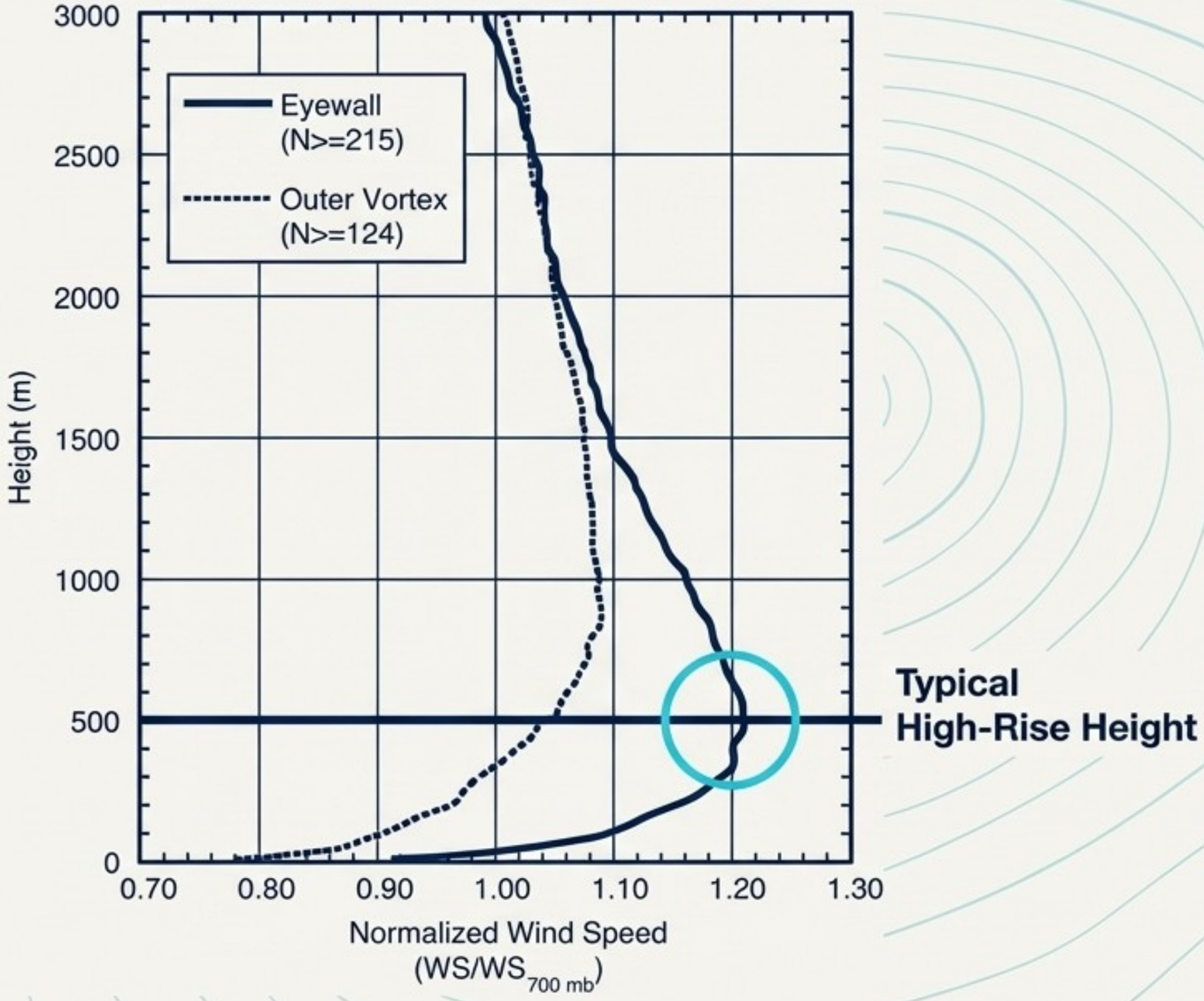
The Low-Level Jet creates supergradient wind loads on high-rises



The LLJ Anomaly

Standard wind profiles assume speed increases steadily with height.

However, in a TCBL, the interaction between radial inflow and eyewall updrafts creates a Low-Level Jet (LLJ).

Winds at ~500m can reach supergradient speeds up to 1.2x faster than the gradient winds above them.

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The governing physics: Navier-Stokes in a cylindrical system

1 Transient evolution over time.

2 Radial, azimuthal, and vertical wind speeds.

3 Vertical diffusivity (turbulence).

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial r} + \frac{v}{r}\frac{\partial u}{\partial \theta} + w\frac{\partial u}{\partial z} - \frac{v^2}{r} = -\frac{1}{\rho}\frac{\partial p}{\partial r} + fv + K\frac{\partial^2 u}{\partial z^2} + \frac{1}{r}\frac{\partial (r\tau_{rz})}{\partial r} + \frac{1}{r}\frac{\partial \tau_{\theta z}}{\partial \theta} + \frac{\partial \tau_{zz}}{\partial z}$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial r} + \frac{v}{r}\frac{\partial v}{\partial \theta} + w\frac{\partial v}{\partial z} + \frac{uv}{r} = -\frac{1}{\rho r}\frac{\partial p}{\partial \theta} - fu + K\frac{\partial^2 v}{\partial z^2} + \frac{1}{r^2}\frac{\partial (r^2\tau_{rz})}{\partial r} + \frac{1}{r}\frac{\partial \tau_{\theta\theta}}{\partial \theta} + \frac{\partial \tau_{z\theta}}{\partial z}$$

1 Transient evolution over time.

2 Radial, azimuthal, and vertical wind speeds.

4 The gradient wind forcing the system from above.

Takeaway: Modeling the TCBL is the art of solving these equations for momentum, heat, and moisture. Doing it perfectly requires supercomputers; doing it efficiently requires clever engineering simplifications.

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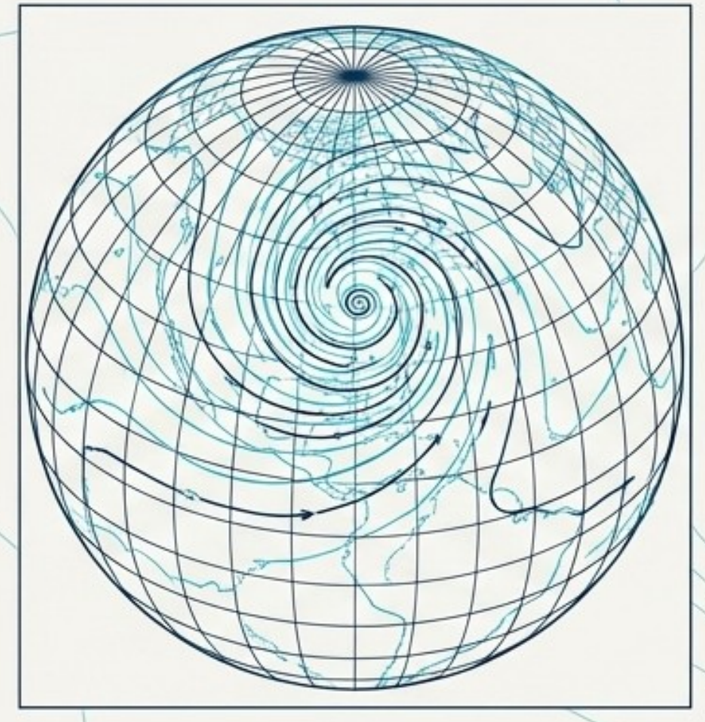
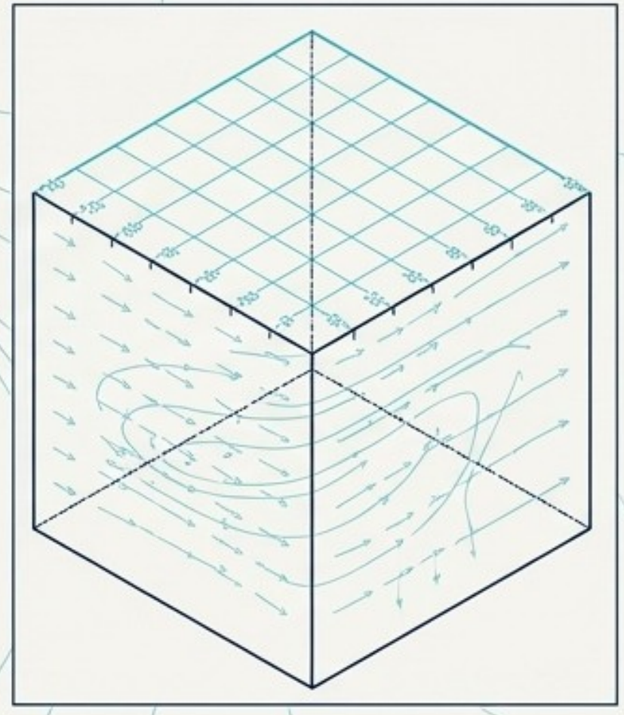
The great divide in wind modeling: Prognostic vs. Diagnostic



Full-Physics / Prognostic

- Solves the primitive equations for the entire atmosphere over time. Tracks transient storm evolution.
- Primary User: Meteorologists forecasting tomorrow's weather.



Diagnostic / Quasi-steady

- Solves for a quasi-steady 3D wind field only within the TCBL, forced by prescribed gradient winds at the top boundary.
- Primary User: Structural engineers running Monte Carlo life-cycle simulations.

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Matrix 1: The Physics-to-Engineering Spectrum

Dimension	Full-Physics (e.g., WRF)	Diagnostic Models
Vertical Domain	Full atmosphere (surface to stratosphere).	Bottom 1-3km only.
Temporal State	Prognostic (tracks evolution).	Quasi-steady (snapshots).
Computational Cost	Massive (Days on computing clusters).	Minute (Seconds to minutes per storm).
Data Assimilation	Requires continuous real-time observational data.	Not required; uses statistical track data.
Monte Carlo Viability	Currently unviable for thousands of runs.	Highly viable; the industry standard.

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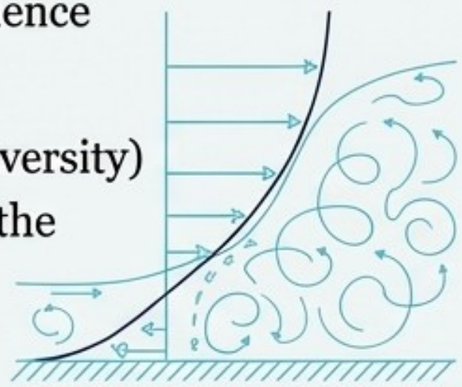
Full-Physics Models: The high-fidelity meteorological standard

WRF-ARW (Advanced Research WRF)

Focuses on the impact of turbulence parameterization schemes.

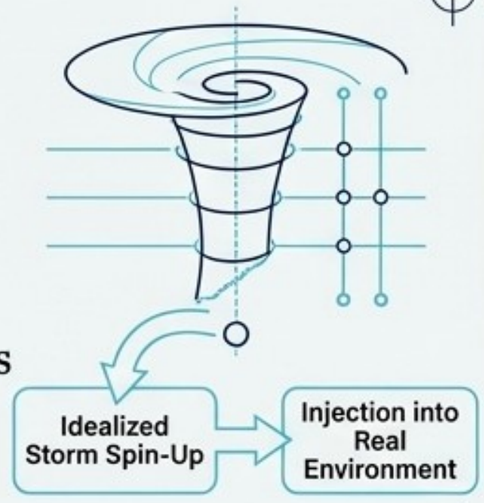
Notes that the YSU (Yonsei University) scheme generally outperforms the MYJ scheme in capturing the boundary layer structure.



HWRF & HWCM

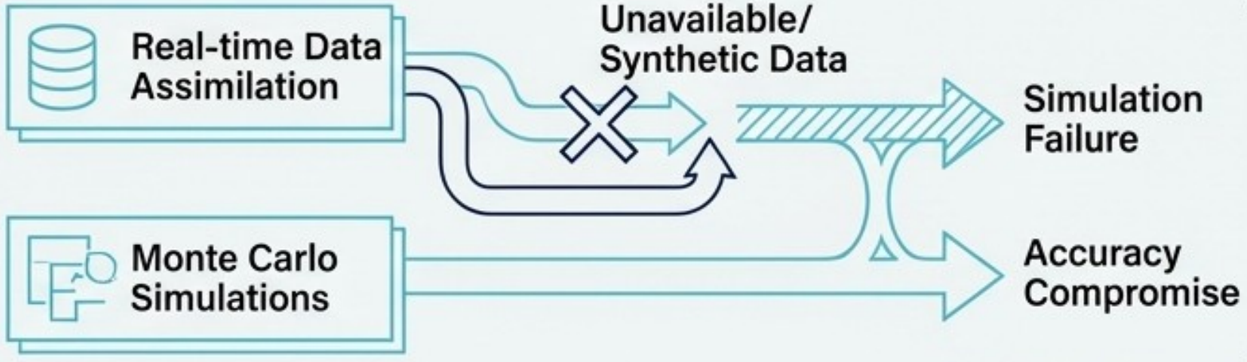
HWRF incorporates advanced vertical diffusivity adjustments based on in-situ measurements.

HWCM (Hybrid WRF Cyclone Model) spins up idealized storms before injecting them into real environments.



The Engineering Hurdle

While highly accurate, these finite-difference NWP models require data assimilation that simply does not exist for the synthetic, generalized storms generated in PBWE Monte Carlo simulations.



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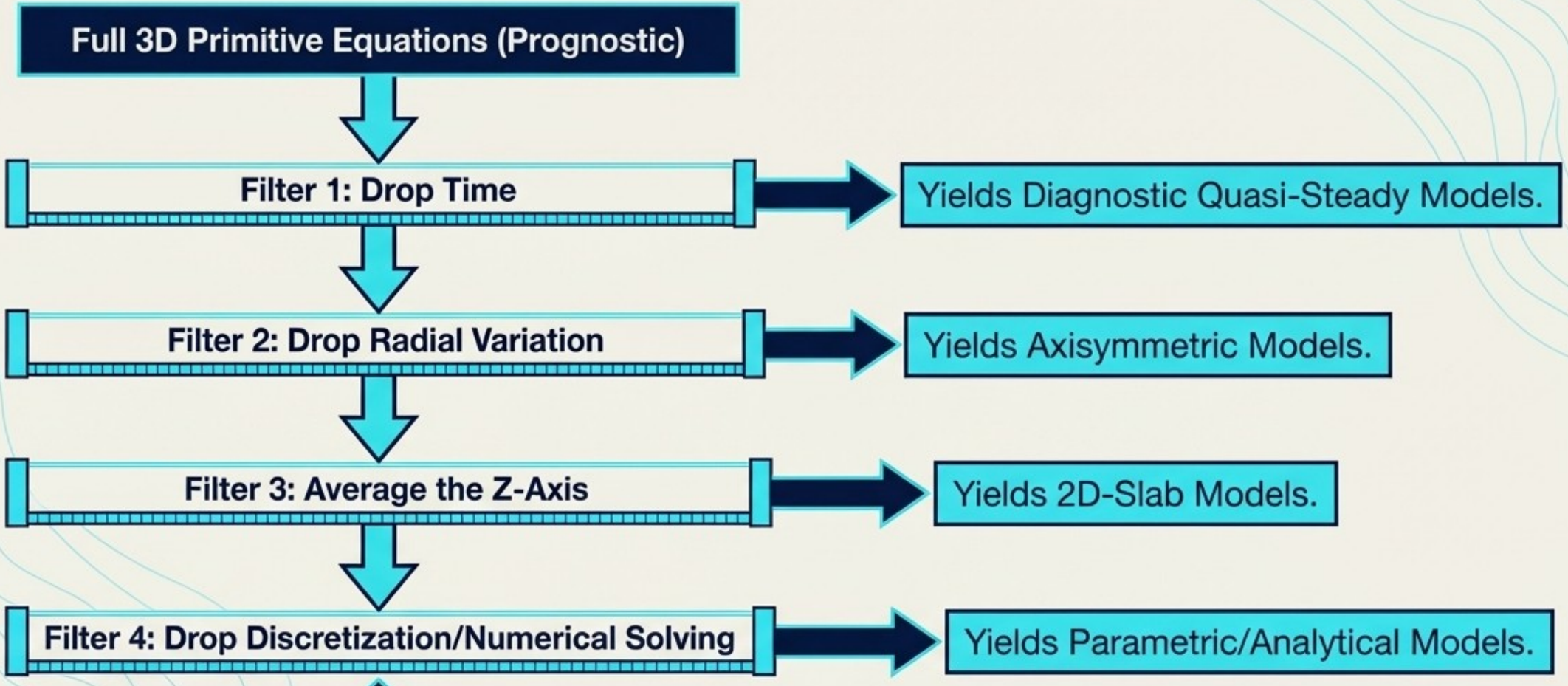
The Diagnostic Simplification Filter



Key Insight: Every diagnostic model trades a piece of mathematical rigor for a massive gain in computational speed.

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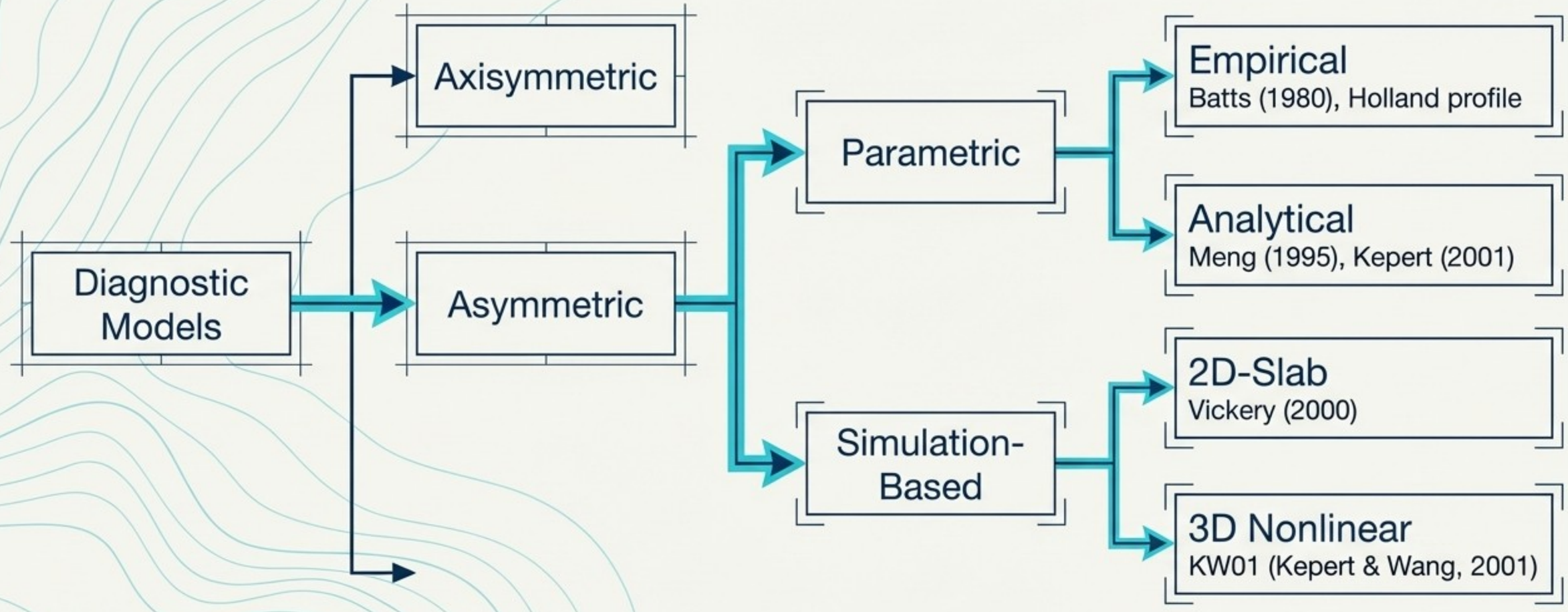
Mapping the Diagnostic Model Hierarchy



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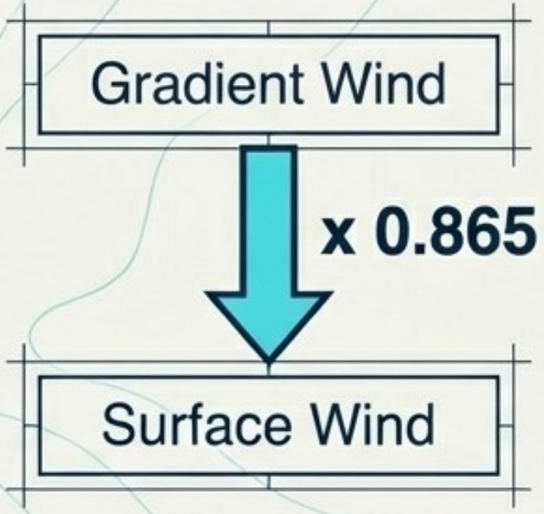
Empirical Approximations: Maximum speed, minimum compute

The SWRF Shortcut



Highlights early methods like Batts (1980) that simply took a gradient wind profile and applied a flat Surface Wind Reduction Factor (SWRF), often exactly 0.865, to estimate surface winds.

Holland's Profile

$$p(r) = p_c + \Delta p \cdot e^{-(A/r^B)}$$

Explains the gradient pressure/wind profile forcing these models. The Holland-B parameter remains crucial today because it requires minimal inputs while highly adapting to match actual reconnaissance pressure profiles, making it perfect for Monte Carlo simulations.

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Analytical Models: Finding the closed-form solution

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\frac{1}{\rho}\nabla\nabla p + f(\mathbf{k}\times\mathbf{u}) + K\nabla^2\mathbf{u}$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\frac{1}{\rho}\nabla\nabla p + f(\mathbf{k}\times\mathbf{u}) + K\nabla^2\mathbf{u}$$

$$+((\mathbf{u}\cdot\nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + f(\mathbf{k}\times\mathbf{u}) + K\nabla^2\mathbf{u}$$

$$+\frac{\partial \mathbf{u}}{\partial t}\cdot\mathbf{b} = -\frac{\partial \mathbf{u}}{\partial t}\cdot(\mathbf{u}\cdot\nabla)\mathbf{S}\cdot\mathbf{u} = \frac{1}{\omega r}\nabla\mathbf{u}$$

$$+(\mathbf{u}\cdot\nabla))\mathbf{u} = -\frac{1}{\rho}\nabla\nabla p + f(\mathbf{k}\times\mathbf{u}) + K\nabla^2\mathbf{u}$$

Linearizing Advection Terms

Approximate Closed-Form Solution

$$u(r,\theta) = F(r)\cdot G(\theta) \approx C\cdot e^{-\lambda r}\cdot\cos(\theta-\alpha)$$

- **Concept:** Instead of numerically simulating a grid, analytical models (like Meng 1995 and Kepert 2001) simplify the primitive equations by linearizing the advection terms.
- **The Result:** This yields an approximate analytical (or semi-analytical) solution that can be calculated instantly.
- **Drawback:** While mathematically elegant, these models often struggle to capture the most severe, highly non-linear dynamics found near the eyewall of a Category 5 hurricane.

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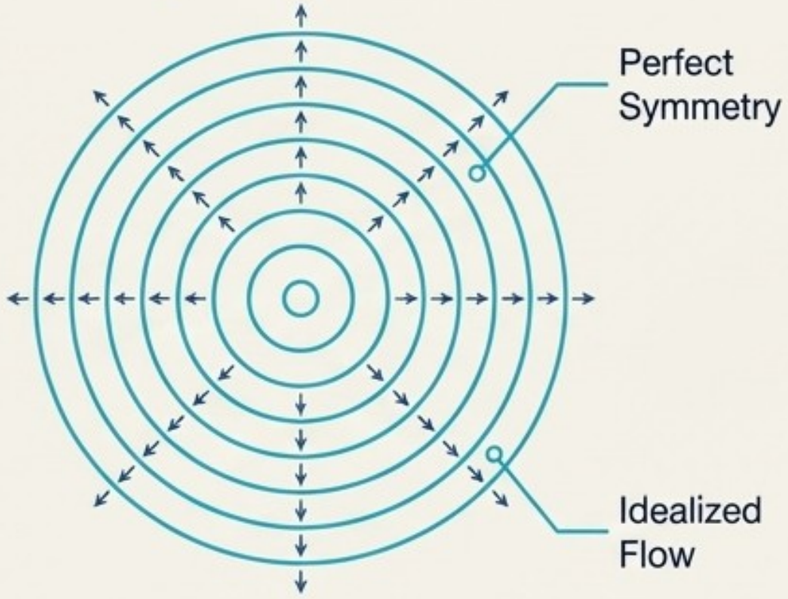
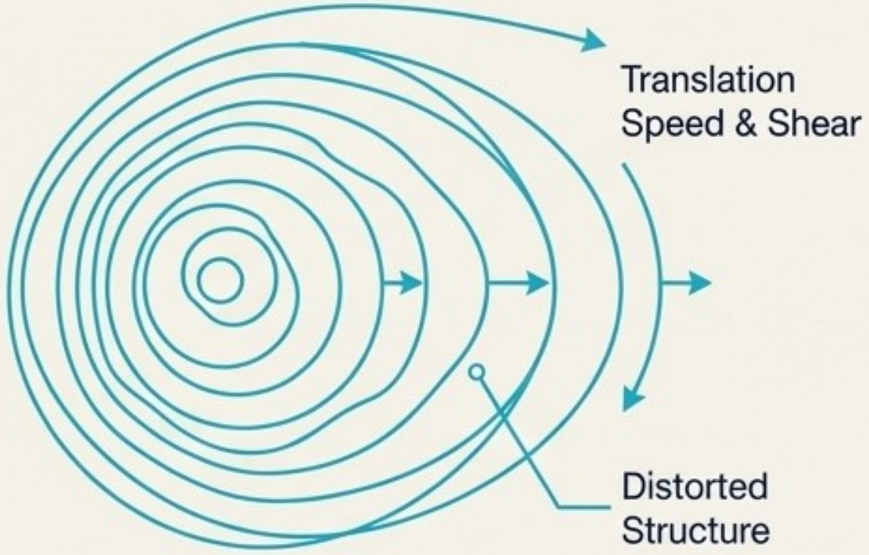
Axisymmetric Models: Stripping away the chaos

Axisymmetric Ideal



Asymmetric Reality

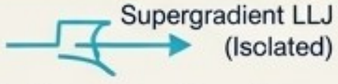


The Math

These models intentionally drop the azimuthal advection terms ($\partial/\partial\phi$) in the primitive equations.

$$\left[u\left(\frac{\partial v}{\partial \phi}\right) + v\left(\frac{\partial u}{\partial \phi}\right) \rightarrow 0 \right]$$

The Utility



Real storms are highly asymmetric due to translation speed and surface drag. Axisymmetric models (like Smith 1968 or Kuo 1982) are idealistic, but they are essential laboratory tools for isolating and proving physical concepts—like confirming that a supergradient Low-Level Jet exists even without storm translation.

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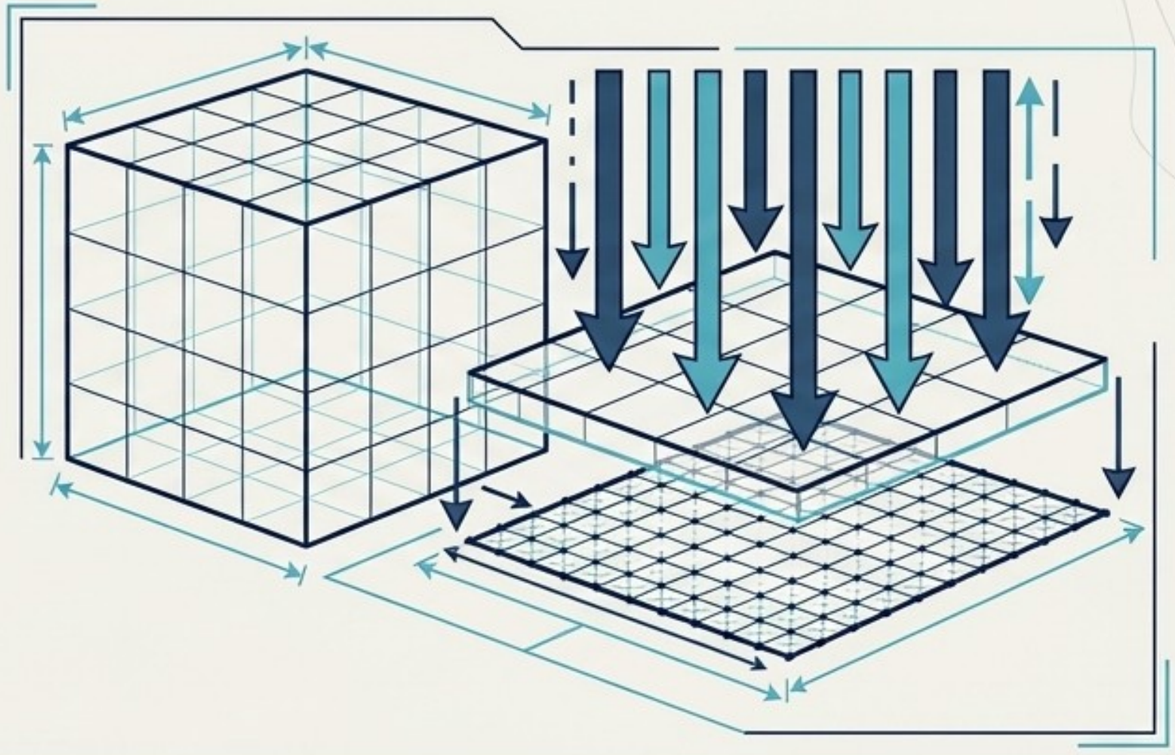
2D-Slab Models: Compressing the Z-Axis



The Mechanism

To save computational power, 2D-Slab models integrate the primitive equations over the entire height of the TCBL.

The Output

They produce height-averaged wind speeds (u_b, v_b).

The Compromise

Because the vertical coordinate (z) is averaged out, these models cannot directly output the vertical wind profile. Engineers must rely on separate empirical profiles to guess what the winds look like at a specific building height.

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Matrix 2: Evaluating the Diagnostic Hierarchy

	Empirical	Analytical	2D-Slab	3D Nonlinear
Mathematical Rigor	Low	Moderate	High	Highest
Asymmetry Handling	SWRF adjustments	Linearized	Dynamically resolved	Dynamically resolved
Vertical Profile	Assumed	Approximated	Averaged out	Explicitly simulated
Primary Drawback	Too simplistic	Fails near eyewall	Misses vertical sheer	Computationally heavy

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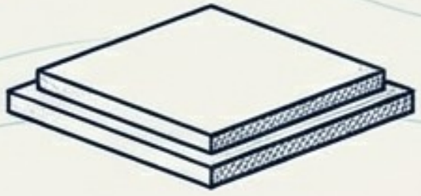
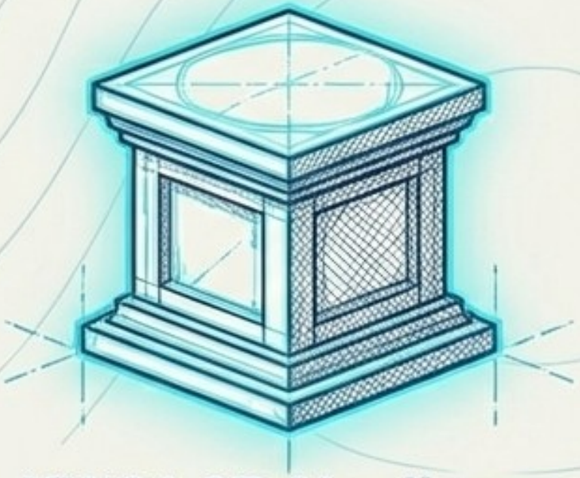
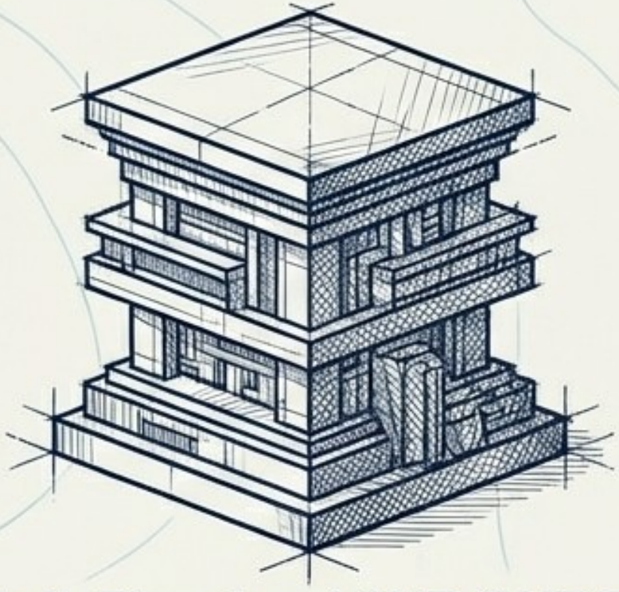
The PBWE Benchmark: The 3D Nonlinear TCBL Model



Empirical Model
Too simple.



KW01 3D Nonlinear
Just right.



Full-Physics NWP (WRF)
Too slow.

The KW01 Model (Kepert & Wang, 2001) occupies the optimum intersection of atmospheric rigor and engineering efficiency. It solves the full 3D diagnostic equations using finite-difference numerical schemes without linearizing the advection terms or averaging the Z-axis.

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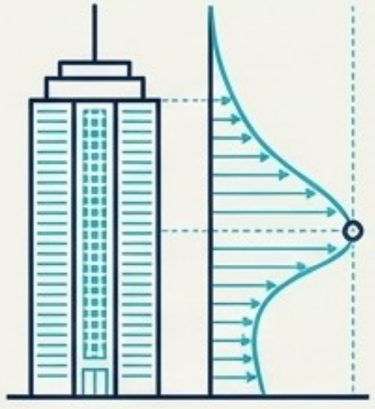
Why KW01 powers the future of structural simulation

Explicit Vertical Resolution

Because it retains the Z-axis, it organically simulates the Low-Level Jet (LLJ) at 500m without relying on generic empirical guesses, ensuring high-rises are designed for actual peak loads.

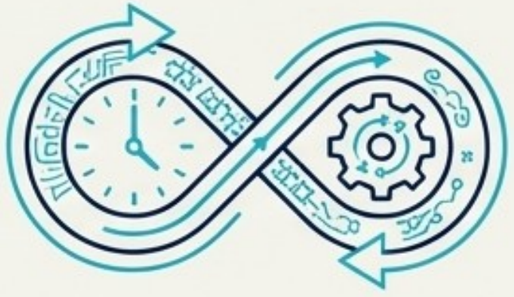


Non-Linear Asymmetry

Accurately captures the lopsided wind fields caused by storm translation speed and changing terrain surface roughness (e.g., ocean vs. city).

Monte Carlo Efficiency

Despite its mathematical rigor, because it is diagnostic and quasi-steady, a single storm can be simulated in seconds—allowing engineers to run the tens of thousands of lifecycles required for true Performance-Based Wind Engineering.



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